INFRINGING VIDEO RETRIEVAL VIA DECOUPLED REPRESENTATIONS AND MUTUAL INFORMATION MAXIMIZATION

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ABSTRACT

Video copyright infringements is a major issue faced by the self-media industry and video sharing platforms, causing serious damages to the interests and enthusiasm of video creators. Identifying such infringing videos is therefore a compelling task. Current state-of-the-art methods typically build on frame-level feature representations extract via pre-trained deep neural networks. Despite its simplicity, this paradigm heavily relies on the original feature representation and lacks an effective constraint on guaranteeing that useful task-relevant semantics are learned from the feature. In this paper, we seek to tackle these challenges from two aspects: (1) We propose to disentangle the high-dimensional original feature Infringing Video Retrieval via Decoupled Representations and Mutual Information Maximizationation into multiple sub-features, explicitly decoupling the feature into exclusive lower-dimensional components. We expect the sub-features to encode non-overlapping semantics of the original feature and reduce task-irrelevant redundancy. (2) Besides the disentangled sub-features, we further learn a supplementary feature, and formulate a mutual information objective to enforce sufficiency of task-relevant information. Extensive experiments on two large-scale benchmark datasets SVD and VCSL demonstrate that our method achieves 88.87% and 94.46% TOP-100 accuracies respectively, significantly surpassing state-of-the-arts. Interestingly, visualization of the extracted features reveals that our approach indeed extracts disentangled sub-features and holistic sufficient feature representation. Our code and model are uploaded in the supplementary file and anonymously released at `https://github.com/iclranonymous/DMI`.

1 INTRODUCTION

The advent of the self-media industry, fueled by the proliferation of video sharing and streaming platforms, has taken the Internet by storm. It is estimated that a person spends an average of 30 minutes per day watching online videos and an appealing and high-quality video on YouTube could attract millions of views in a single day. The sheer skyrocketing of video media on multiple Internet platforms has also brought along many issues such as copyright infringement, which is a major concern for artists, authors, and publishers, who have to spend numerous resources to prevent plagiarism. As a result, Infringing Video Retrieval (IVR) has become an increasingly pertinent task and has drawn increasing attention from the research community.

Infringing video retrieval amounts to detecting similar and near-duplicate video pairs from a video gallery. Over the past decade, (Cai et al., 2011; Kordopatis-Zilos et al., 2017c; Liao et al., 2018) and hashing functions (Song et al., 2011; 2013) have been popular solutions to generate video representations and enable similarity computation. However, these models heavily depend on handcrafted features and shallow functions to identify infringements, leading to unsatisfactory accuracy. Recently, deep learning models have become the dominant approaches in infringing video retrieval, by virtue of their capacity to learn non-linear functions and sophisticated feature representations. One line of works propose to fuse all the frame features into a single video-level representation and perform similarity matching on such representations (Liu et al., 2013; Kordopatis-Zilos et al., 2017b; Liu et al., 2017). Another line of works advocate to compute a frame-to-frame similarity ma-

trix for all the frame-level features, which are subsequently combined into a video-level similarity score (Tan et al., 2009; Hu & Lu, 2018; Baraldi et al., 2018).

Fundamentally, the essence of infringing video retrieval amounts to obtaining feature representations for the input videos pair which can essentially capture their content information and disregard superficial differences (such as varying video encodings, noises or pixel-level differences) for effective discriminations. Upon investigating and experimenting on the released implementations of state-of-the-art approaches (Kordopatis-Zilos et al., 2017b; Liu et al., 2017), we empirically observe that two issues still persist: (1) Although extensive experiments confirm the efficacy of existing methods in retrieving near-duplicate video pairs, current methods are shown to have difficulties in identifying complicated cases where two videos are not directly similar, *e.g.*, an infringing video might be obtained by cutting out part of the picture, video splicing, blocking part of the frames etc. (2) Current state-of-the-art methods rely on neural networks to extract task-relevant semantics from the original feature, lacking mathematical constraints guaranteeing the reduction of superfluous or redundant information. Consequently, the final learned representations may contain superficial differences, which compromises robustness and performance of such methods.

In this paper, we try to tackle these challenges by embracing two key designs. (i) Following theoretical analysis steeped in the framework of contrastive learning and Kullback–Leibler divergence, we first explicitly disentangle the original backbone feature into a group of non-overlapping sub-features. (ii) On top of the sub-features, we engage in mutual information theory to further extract supplementary feature from the original input. The sub-features and the supplementary feature together are composed into the final feature. We attempt to maximize the mutual information between the label and the supplementary feature to ensure sufficiency of the final feature representation, *i.e.* no task relevant information is dropped. The framework follows a maximum relevancy and minimum redundancy paradigm, serving to remove superfluous information and retaining as much important relevant information as possible for the end task. Our proposed procedure constitutes an effective knowledge distillation procedure that explicitly disentangles the high-dimensional original feature and helps in distilling task-relevant cues from the original input features. We also visualized the decoupled sub-features, investigating the feasibility of the framework.

We extensively evaluate the proposed method on two large-scale benchmark datasets, SVD and VCSL. Empirical evaluations show that our approach significantly outperforms current state-of-the-art methods. Our method achieves 88.87% and 94.46% TOP-100 accuracies on SVD and VCSL, respectively. Our code is open-sourced on GitHub at `https://github.com/iclranonymous/DMI` (anonymized). We also present extensive ablation analysis on the contribution of each component, and validate the efficacy of feature decoupling and the proposed mutual information loss.

To summarize, our contributions are:

- We introduce a novel framework for extracting intermediate sub-features, which incorporates KL divergence maximization and contrastive learning to supervise the decoupling of sub-features. The disentangled sub-features each serves to preserve a unique aspect of the original input feature that is closely-related to the task.
- In order to effectively extract task relevant information missed by the sub-features, we systematically propose a supplementary feature extraction method entrenched in the mutual information theory, which seeks to distill as much task relevant information from the original feature as possible.
- Extensive experiments on the SVD and VCSL benchmark datasets show that our method surpasses state-of-the-art approaches. Interesting findings and insights on original backbone feature selection, sub-features, and the final feature are presented.

2 Related Work

Before diving into the details of our approach, we briefly provide a holistic overview on the literature that is closely pertinent to this work. The closely-related literature can be roughly cast into three topics, namely *video retrieval*, *contrastive learning*, and *mutual information theory*.

2.1 VIDEO RETRIEVAL

Earlier approaches for video retrieval mainly revolve around code books (Cai et al., 2011; Kordopatis-Zilos et al., 2017c; Liao et al., 2018) and hashing functions (Song et al., 2011; 2013) for encoding a video to a low-dimensional representation. Fueled by the success of deep learning in recent years, the predominant approach is to decompose the video as frames, which are then fed into an image extraction backbone network to obtain a sequence of image feature representations. One breed of research fuses all these image features into a single video-level representation and perform matching on video-level representations (Liu et al., 2013; Kordopatis-Zilos et al., 2017b; Liu et al., 2017). Another line of work computes a frame-to-frame similarity matrix for all the frame-level features, which are subsequently combined into a video-level similarity score (Tan et al., 2009; Hu & Lu, 2018; Baraldi et al., 2018). However, current methods heavily rely on the network architecture to obtain effective features from the input videos pair, with no explicit mathematical constraints that could be leveraged to enforce extraction of useful semantic cues and removal of redundancies. Our work seeks to address this gap by proposing an effective knowledge distillation process that facilitates the video retrieval task.

2.2 CONTRASTIVE LEARNING

Contrastive learning, which is typically done in a self-supervised manner, seeks to learn a representation that forces dissimilar data samples to be far apart in the encoding space while similar data samples are close to each other. One seminal work (Oord et al., 2018) proposes the InfoNCE loss as an effective contrastive learning objective. Furthermore, (Chen et al., 2020a) propose SimCLR, which engage data augmentations to generate positive pairs. Another landmark work, MoCo (He et al., 2020), improves upon the InfoNCE work through the use of a momentum contrast mechanism that improves convergence. (Yokoo, 2021) further advocates employing convolutional neural networks with contrastive learning and hard data augmentation, which is empirically effective for obtaining more discriminative representations. Both SimCLR (Chen et al., 2020a) and (Yokoo, 2021) employ a similar strategy of data augmentation, where the augmented image would serve as a positive sample. Such data augmentations include cropping, grayscale, blocking part of the picture, and horizontal flipping, which are exactly the same common practices appearing in video infringement. As such, we employ the MoCoV2 (Chen et al., 2020b) and ISC (Yokoo, 2021) pre-trained models as backbone networks for our base feature extraction.

2.3 MUTUAL INFORMATION THEORY

The mutual information of two random variables gives a quantitative measurement of the mutual dependency between them. The seminal work (Tishby et al., 2000; Tishby & Zaslavsky, 2015) of information bottleneck theory puts forth a theory of representation learning, using the analogy of a bottleneck process that squeezes out redundant and irrelevant input noises, retaining only important information that may be utilized for the task. Our work may be readily understood within such a framework, whereby our proposed method processes the high dimensional backbone extracted base feature to remove superfluous information, arriving at a compact low dimensional feature representation that is more robust and captures the essence of similarity comparison for videos. Inspired by (Tishby & Zaslavsky, 2015; Liu et al., 2022; Tian et al., 2021), we similarly invoke mutual information to perform feature level supervision to perform knowledge distillation.

3 METHOD

Problem Presented with a query video and a video gallery, we are interested in retrieving all similar videos from the video gallery. Specifically, we seek to learn an embedding for each video, where the similarity of a video pair can be conveniently computed via a distance function on their respective embedding vectors. In other words, our task can be deemed as a supervised metric learning problem where the effectiveness of the embedding is assessed based on whether similar videos have a correspondingly smaller distance in the embedding space. Unsurprisingly, the embedding vectors for non-similar videos should be far apart and have a low similarity score.

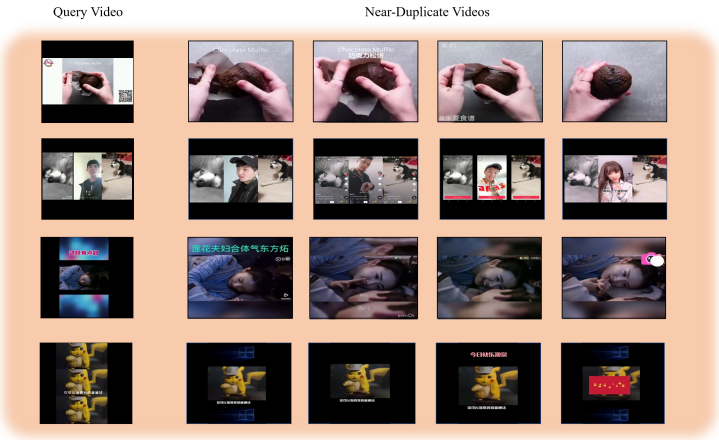


Figure 1: **Examples from SVD dataset**: Video infringement typically involves editing, splicing, blocking, etc.

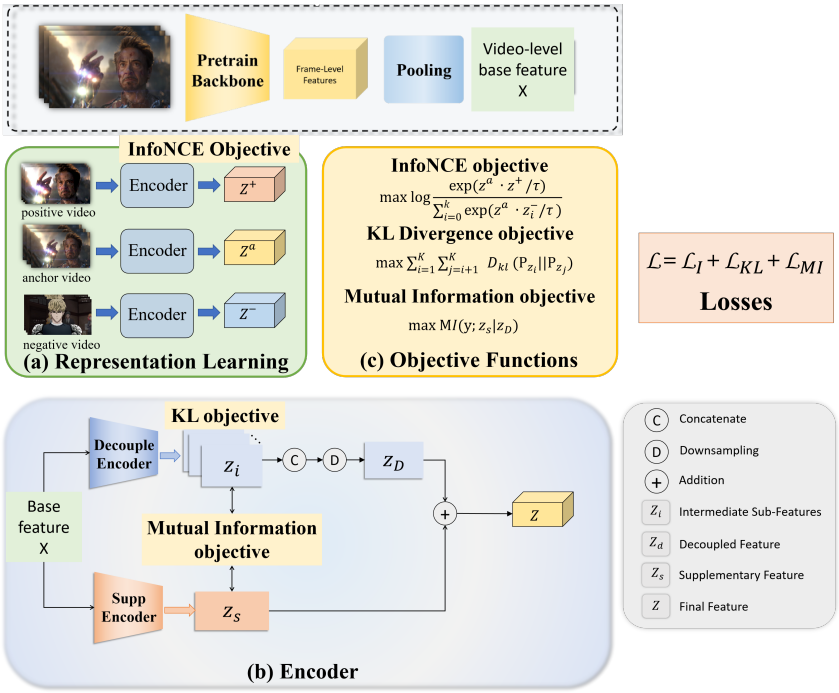


Figure 2: **Our framework**: A base video-level feature is extracted following a pre-trained backbone network and average pooling. Subsequently, we extract disentangled intermediate sub-features and a supplementary sub-feature to prevent possible relevant information dropping, before finally fusing these sub-features into a final feature. This final feature serves as the learned video representation which will be used in the similarity retrieval task.

3.1 Method Overview

An overview of our framework is presented in Figure 2. First, we employ a pre-trained network, such as MoCoV2 (He et al., 2020), as the backbone network to obtain a sequence of frame-level features for each input video. Subsequently, we fuse all frame-level features through average pooling, arriving at a video-level base feature x.

Given the base feature representation x, **(1)** we propose a disentangled sub-features extraction module, which extracts a total of k decoupled intermediate sub-features $\mathbf{z}_i$. The key component in this module is to maximize the distributional distance (such as the KL divergence and Wasserstein distance) between each pair of sub-features. We expect each sub-feature to encode exclusive aspects of the original video feature. Such a decoupling would be akin to a divide-and-conquer strategy, where we break the representation into lower dimensional disentangled components to remove redundancy and facilitate the extraction of task-relevant information. **(2)** In order to account for possible task relevant information loss in the decoupled representations $\mathbf{z}_i$ and ensure the sufficiency of the encoded representation, we further introduce another supplementary feature $\mathbf{z}_s$. We leverage a mutual information objective to effectuate task-relevant supplementary information extraction. The decoupled feature $\mathbf{z}_d$ and the supplementary feature $\mathbf{z}_s$ are then summed to arrive at our final feature representation.

On top the disentanglement and mutual information objectives above, we also incorporate the *infoNCE loss* for contrastive learning, to enforce similar videos to have closely-clustered embeddings and dissimilar videos to have distinct embeddings. In what follows, we will elaborate our key components of disentangled sub-feature learning, mutual information based supplementary information learning, and contrastive learning respectively.

3.2 Disentangled Sub-features Extraction Module

We can certainly train a direct end-to-end CNN network for extracting infringement-related video features, as is done in most previous methods (Liu et al., 2013; Kordopatis-Zilos et al., 2017b; Liu et al., 2017). However, such approaches implicitly mix different kinds of video features together, translating to high feature redundancy and low interpretability. Inspired by this, we believe it would be fruitful to investigate whether introducing supervision to disentangle the features would facilitate the task. With this goal in mind, we design a sub-features extraction module by maximizing the Kullback–Leibler divergence, which maps the original d-dimensional video feature x to k disentangled lower dimensional sub-features $\{\mathbf{z}_i\}_{i=1}^{k}$. The KL divergence objective constrains the sub-features to each encode minimally-overlapping semantics of the input video feature. More specifically, we maximize the KL divergence between all sub-feature encodings, which can be formulated as:

$$\max \sum_{i=1}^{k}\sum_{j=i+1}^{k} \mathcal{D}_{KL}(\mathbb{P}_{\mathbf{z}_i}||\mathbb{P}_{z_j}) = \max\Big[-\sum_{i=1}^{k}\sum_{j=i+1}^{k}\int \mathbb{P}_{\mathbf{z}_i}(x) ln(\frac{\mathbb{P}_{z_j}(x)}{\mathbb{P}_{\mathbf{z}_i}(x)}) d.x\Big] \tag{1}$$

Through such a principled decoupling of the input high dimensional feature, we remove task-irrelevant redundant information and obtain compact disentangled sub-features representation which serves to facilitate subsequent steps of task-relevant information extraction.

3.3 Supplementary Feature Encoding via Mutual Information

Up to this point, we have acquired a disentangled encoding of the base feature. However, in this process, the disentangled sub-features might also omit some useful information from the base feature, leading to insufficiency and erosion of task-relevant information. Therefore, we propose to complement the decoupled feature representations $\mathbf{z}_i$ with an additional feature $\mathbf{z}_s$ that consists of missing task-relevant information. To achieve this goal, we theoretically analyzed the mutual information between the label and the features, and derive a loss that maximizes the extraction of complementary task-relevant features. This allows us to enhance $\mathbf{z}_i$ and achieve a final feature that is a sufficient representation of the base feature with respect to the task.

Formally, mutual information is a measure of the mutual dependency between random variables. The mutual information $\mathcal{I}(\mathbf{x_1};\mathbf{x_2})$ between two random variables $\mathbf{x_1}$ and $\mathbf{x_2}$ quantifies their correlated information and is defined as:

$$\mathcal{I}(\mathbf{x_1};\mathbf{x_2}) = \mathbb{E}_{\mathbf{p}(\mathbf{x_1},\mathbf{x_2})}\left[\log \frac{\mathbf{p}(\mathbf{x_1},\mathbf{x_2})}{\mathbf{p}(\mathbf{x_1})\mathbf{p}(\mathbf{x_2})}\right], \tag{2}$$

where $p(\mathbf{x_1},\mathbf{x_2})$ is the joint probability distribution of $\mathbf{x_1}$ and $\mathbf{x_2}$, while $p(\mathbf{x_1})$ and $p(\mathbf{x_2})$ are their marginals.

Given the acquired sub-features $\mathbf{z} = \langle \mathbf{z}_1, \mathbf{z}_2, \cdots, \mathbf{z}_k \rangle$, our primary objective is to maximize the amount of complementary task-relevant information in the supplementary feature $\mathbf{z}_s$:

$$\max \ \mathcal{I}(y; \mathbf{z}_s | \mathbf{z}) \tag{3}$$

where y denotes the label, and $\mathcal{I}(y; \mathbf{z}_s|\mathbf{z})$ represents the amount of task-relevant information in the supplementary feature $\mathbf{z}_s$. Intuitively, maximizing this objective amounts to optimizing the additional relevant and complementary information we seek to extract from the original feature.

To proceed, we factorize Eq. (3) as below:

$$\max \ \mathcal{I}(y; \mathbf{z}_s|\mathbf{z}) \Rightarrow \max \ [\mathcal{I}(y; \mathbf{z}_s) - \mathcal{I}(\mathbf{z}_s; \mathbf{z}) + \mathcal{I}(\mathbf{z}_s; \mathbf{z}|y)] \tag{4}$$

where $\mathcal{I}(\mathbf{z}_s; \mathbf{z}|y)$ represents the overlapping task-irrelevant information between $\mathbf{z}_s$ and $\mathbf{z}$. We may assume that the task-irrelevant information shared between $\mathbf{z}_s$ and $\mathbf{z}$ will be negligible upon sufficient training (Tian et al., 2021). This simplifies Eq. (4) to:

$$\max \ \mathcal{I}(y; \mathbf{z}_s|\mathbf{z}) \longrightarrow \max \ [\mathcal{I}(y; \mathbf{z}_s) - \mathcal{I}(\mathbf{z}_s; \mathbf{z})] \tag{5}$$

To prevent information dropping, we further introduce a regularization term that is formulated as:

$$\min \sum_{i=1}^{k} \mathcal{I}(y; \mathbf{z}_i|\mathbf{z}_s) \tag{6}$$

where $\mathcal{I}(y; \mathbf{z}_i|\mathbf{z}_s)$ measures the vanishing task-relevant information in $\mathbf{z}_i$ during feature disentangling. Analogous to the procedure from Eq. (4) to Eq. (5), we simplify the regularization term in Eq. (6) as follows:

$$\min \ \sum_{i=1}^{k} \mathcal{I}(y; \mathbf{z}_i|\mathbf{z}_s) \longrightarrow \min \ \sum_{i=0}^{k} [\mathcal{I}(y; \mathbf{z}_i) - \mathcal{I}(\mathbf{z}_i; \mathbf{z}_s)]. \tag{7}$$

Finally, we arrive at our mutual information objective:

$$\max \mathcal{I}(y; \mathbf{z}_s|\mathbf{z}) + \min \sum_{i=0}^{k} \mathcal{I}(y; \mathbf{z}_i|\mathbf{z}_s) + \min \sum_{i=1}^{k} \mathcal{I}(y; \mathbf{z}_i|\mathbf{z}_s) \tag{8}$$

3.4 Contrastive learning via InfoNCE loss

Finally, we concatenate all $\mathbf{z}_i$ and perform a down-sampling to obtain a single decoupled feature $\mathbf{z}_d$. This is then summed with the supplementary feature $\mathbf{z}_s$, giving our final feature representation $\mathbf{z}$. We then supervise this loss via the InfoNCE loss in a contrastive learning framework.

The intuition behind contrastive learning is to cast videos into embedding vectors where similar videos lie close to each other in the embedding space while dissimilar videos stay away from one another. Given a video v, we denote a positive video v^+ as one that is similar to v and a set of negative video $\{v_j^-\}_{j=1}^N$ that is dissimilar to v. We denote their features as $\mathbf{z}^+$ and $\{\mathbf{z}_j^-\}_{j=1}^N$, respectively. We employ infoNCE loss (Oord et al., 2018), which is given by:

$$L_{\text{infoNCE}} = -log \frac{exp(\mathbf{z} \cdot \mathbf{z}^+/\tau)}{\sum_{j=1}^{N} exp(\mathbf{z} \cdot \mathbf{z}_j^-/\tau)} \tag{9}$$

where τ is a regulation parameter.

4 Experiments

In this section, we evaluate the proposed approach on two large-scale benchmark datasets, SVD and VCSL. We seek to answer the following research questions.

- RQ1: How is the proposed method comparing to state-of-the-art video retrieval approaches?
- RQ2: How much do different components of our method contribute to its performance?
- RQ3: What interesting insights and findings can we obtain from the empirical results?

Next, we first present the experimental settings, followed by answering the research questions one by one.

Table 1: Comparison of our method with previous methods on the SVD dataset

Method	Embedding Dim	Top-100 mAP
DML(Kordopatis-Zilos et al., 2017b)	500D	0.813
CNN-L(Kordopatis-Zilos et al., 2017a)	4096D	0.610
CNN-V	4096D	0.251
SVRTN(He et al., 2021)	512bits	0.871
Our Method	**256D**	**0.889**
MoCoV2 + CS(cosine similarity)	2048D	0.804
VGG + CS	4096D	0.792
MoCoV2 + FC(fully connected)	256D	0.852
VGG + FC	256D	0.830
MoCoV2 + ED(euclidean distance)	2048D	0.799
VGG + ED	4096D	0.785

4.1 Experimental Settings

Datasets The **SVD** Jiang et al. (2019) dataset is a large-scale benchmark dataset for infringing video retrieval. The dataset contains over 500,000 videos and over 30,000 labeled pairs of similar videos. The categories of videos are all-encompassing, including movies, TV series, commentary videos, and daily videos. The **VCSL** He et al. (2022) dataset consists of 160,000 video copy pairs with diverse video categories and varying video durations.

Implementation Details Our DMI (feature Disentanglement and Mutual Information maximization) framework is implemented with PyTorch. For pre-processing, we divide the videos into frames and fed them into pre-trained networks including MoCoV2 Chen et al. (2020b), ISC Yokoo (2021), and VGG Simonyan & Zisserman (2014) backbone networks to extracts the feature representation for each frame. The unlabeled videos are deemed as dissimilar to all other videos and serve as negative training samples during the training process, which improves the robustness of our model. Training is done with 4 Nvidia Geforce RTX 2080 Ti GPUs. The batch sizes on SVD and VCSL datasets are set to 64 and 16. In contrastive learning, the ratio of positive samples to negative samples is 64:2048. All training process is terminated within 300 epochs. The number of sub-features is set to $k = 4$. For evaluation, the widely used top-100 mAP Jiang et al. (2019) metric is adopted.

4.2 Comparisons with State-of-the-art Methods (RQ1)

First, we benchmark our approach against state-of-the-art methods on the SVD dataset. The performance of different methods are presented in terms of Top-100 mAP in Table 1. A total of 10 methods are compared, including DML (Kordopatis-Zilos et al., 2017b), CNN-L (Kordopatis-Zilos et al., 2017a), CNN-V (Kordopatis-Zilos et al., 2017a), and SVRTN (He et al., 2021). Specifically, (1) DML proposes to early or late fuse frame-level features into a single video vector, which is then fine-tuned by deep metric learning. (2) CNN-L & CNN-V advocates to convert intermediate CNN features into one vector via layer and vector aggregation schemes, respectively. (3) SVRTN exploits a transformer structure to aggregate frame-level features into clip-level features and learn the discriminative information from the interactions among clip frames. We further engage in cosine similarity-based and Euclidean distance-based approaches, which respectively employ cosine similarity and Euclidean distance between the original video features to identify similar videos. A baseline network that consists of only fully connected layers to process the original feature is also included in the comparison.

From the Table 1, we observe that metric learning-based and transformer based approaches, namely DML and SVRTN, achieve the current state-of-the-art performance. Our method is able to outperform them with a 7.6% and 1.8% mAP gain, and overall achieves a 88.9% TOP-100 mAP. The quantitative comparisons suggests the feasibility of the proposed approach. Further, we also conduct evaluations on the VCSL dataset. The quantitative results are reported in Table 2:

From the results, we may observe that existing methods are already impressive where the current state-of-the-art method ISCYokoo (2021) obtains a 93.2% mAP. Our method achieves a 94.5% mAP

Method	Embedding Dim	Top-100 mAP
ISC(Yokoo, 2021) + CS	256D	0.926
ISC + FC	256D	0.932
ISC + CS	256D	0.919
MocoV2 + CS	2048D	0.918
MoCoV2 + FC	256D	0.922
MocoV2 + ED	2048D	0.908
Our Method	**256D**	**0.945**

Table 2: Comparison of our method with previous methods on the VCSL dataset

Method	Decouple	Supplementary	top-100 mAP	top-inf mAP
MoCoV2			0.852	0.838
(a)	$\surd$		0.869	0.852
(b)		$\surd$	0.871	0.856
(c)	$\surd$	$\surd$	0.889	0.866
VGG			0.830	0.809
(a)	$\surd$		0.835	0.810
(b)		$\surd$	0.836	0.815
(c)	$\surd$	$\surd$	0.843	0.818

Table 3: Ablation of different components in our method for SVD.

and the 1.3% mAP improvement over the already impressive state-of-the-art provides empirical support for the effectiveness of the proposed method.

4.3 Removal of Sub-features Module and Supplementary Feature Module

By default, the proposed sub-features extraction module and the supplementary feature encoding module are coupled together to approach a high performance. We are interested in studying the impact of removing each module respectively from the proposed method. Experiments are conducted on both SVD dataset and VCSL (He et al., 2022) dataset.

More specifically, we empirically evaluate the efficacy of the sub-features extraction module and the supplementary feature encoding module in our framework. **(a)** For the first setting, we remove the sub-features extraction module in our network, and only use a single feature encoding module to extract features. **(b)** For the second setting, we remove the supplementary feature extraction module in the network, and only adopt the sub-features extraction module to learn representations.

We compare them with the default network. Experimental results on SVD and VCCSL datasets are respectively illustrated in Table 3 and Table 4, where both top-100 mAP and top-inf mAP results are reported. We see that without the sub-features extraction module, the proposed method undergoes a 1.25% and 0.55% mAP drops on the two datasets respectively. These empirical evidences indicate the importance of the sub-features extraction module, which contributes in removing feature redundancy and disentangling features. Meanwhile, without the supplementary feature extraction module, the mAP drops are 1.4% and 0.5% on the two datasets respectively. This suggests the significance of the supplementary feature extraction module, which contributes in capturing more complete task-relevant information.

4.4 Findings

Our first finding is that video infringement usually involves cropping, blocking part of the picture, splicing, and content editing. Figure 1 demonstrates several video infringement examples, where the left most column shows the original video and the right columns show the modified videos.

Our second insights concern the separability between sub-features. We randomly selected 3,750 videos and visualized their extracted four kinds of sub-features $\{\mathbf{z}_i\}_{i=1}^{4}$. The results are depicted in Figure 3. Interestingly, we observe that the four kinds of sub-features are indeed disentangled.

Method	Decouple	Supplementary	top-100 mAP	top-inf mAP
ISC			0.932	0.881
(a)	√		0.937	0.888
(b)		√	0.940	0.885
(c)	√	√	0.945	0.891
MoCoV2			0.922	0.838
(a)	√		0.929	0.865
(b)		√	0.926	0.868
(c)	√	√	0.932	0.871

Table 4: Ablation of different components in our method for VCSL He et al. (2022).

Moreover, besides the MoCoV2 and ISC features, we also tried using VGG-4,096 as the backbone network for extracting original video feature. Upon the obtained VGG feature, we utilized the same proposed pipeline to compute the similarity between videos. Comparisons on the Top-100 mAP results between VGG feature and MoCoV2 feature are illustrated in Table 1. We empirically observe that VGG feature yields much lower accuracy. We conjecture the reasons are that MoCoV2 and ISC backbone network inherently considered data augmentations such as cropping, grayscale, blocking part of the picture, and horizontal flipping, which are exactly the same common practices adopted in video infringement. During the experiment, some interesting phenomena caught our attention. We found the sub-features which extracted by the disentangling module also have a high performance which is reported in Table 5. We also visualize the sub-features in Figure 3.

	MoCoV2		**VGG**	
Features	top-100 mAP	top-inf mAP	top-100 mAP	top-inf mAP
Sub Feature 1	0.863	0.835	0.816	0.795
Sub Feature 2	0.854	0.828	0.815	0.791
Sub Feature 3	0.867	0.839	0.785	0.758
Sub Feature 4	0.852	0.833	0.804	0.777

Table 5: As shown in the table, our sub-features keep a good performance after decouple.

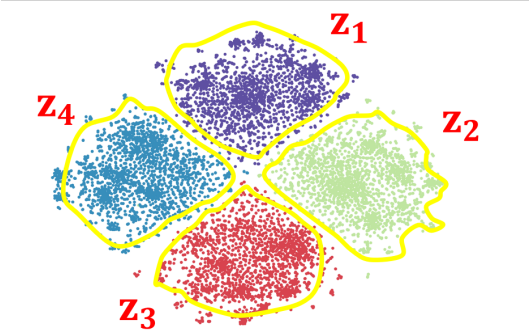


Figure 3: Visualization of our feature decoupling network on 3,750 randomly selected videos.

5 Conclusion

In this work, we have proposed a novel framework for infringing video retrieval task, which is equipped with feature disentanglement and mutual information maximization. Within the framework, we combine contrastive learning and the KL divergence maximization to effectively supervise the decoupling of sub-features. Further, we theoretically derive a mutual information objective for supervising the extraction of supplementary and task-relevant features that might be missed by sub-features. Extensive experiments demonstrate that our method achieves state-of-the-art results on two benchmark datasets, SVD and VCSL.

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